



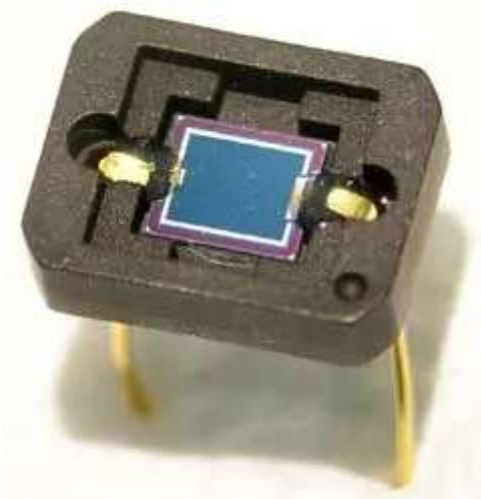
Photodiode Basics

The photodiode is a very fast acting photo detector which produces a very small output current that can be converted into a voltage when exposed to light

Optoelectronic Devices

The **Photodiode** uses a pn-semiconductor junction to convert surrounding light energy into an electrical signal which can then be processed and used to monitor some physical condition. As such, photodiodes are a fundamental component in the branch of electronics known more commonly as “Optoelectronics”. Opto-electronic devices, such as photo-diodes, photo-transistors, and LDR’s are able to source, detect, and control light.

Optoelectronic devices convert electrical signals into optical signals and vice versa. As such, we can combine both optical and electronic solid-state technologies together to focus on the generation, signal processing, switching, amplification, and detection of light signals across a spectrum of wavelengths.



Optoelectronic devices, which includes photodiodes, have applications in everything from internet communications through fibre optics to digital cameras. As well as in your mobile device, smart watches, fitness trackers, and smoke detectors.

They can be used as either individual components, in photodiode arrays, or as photonic integrated circuits for use in LED/OLED video displays, opto-isolators, solar panels, LED lighting systems, etc, etc.

We can therefore create a list of the main optoelectronic components which could be use in a particular project as being:

Light Generating Sources:

- Light-Emitting Diodes (LEDs) – Convert electrical current directly into light through electroluminescence
- Liquid Crystal Displays (LCDs) – Modulate light transmission using liquid crystals
- Laser Diodes (LDs) – Produce coherent, monochromatic light through stimulated emission

- Organic LEDs (OLEDs) – Use organic compounds for light emission of displays

Light Detecting Devices

- Photodiodes – Are photovoltaic detectors which convert photon light into an electrical current
- Phototransistors – Provide current amplification of detected light signals
- Avalanche Photodiodes (APDs) – Offer high sensitivity through internal gain
- Charge-Coupled Devices (CCDs) – Convert light patterns into digital signals

Then we can see that opto-emitters and opto-detectors (photovoltaic detectors) either generate light or use light during their operation. Whether that is as a sensor, LED lighting or display. But the optoelectronic photo-device we are interested in here are the “Photodiodes”.

Photodiode Basics

The two-layer [PN-junction semiconductor diode](#) has many uses in the field of electrical and electronic engineering. They can be used in rectifiers, voltage regulation and wave-shaping circuits. But they are also extremely useful as optoelectronic devices in which one type of diode allows light to control its current. While another type of diode emits light when passing a current. These two devices are the **Photodiode** and the **Light Emitting Diode** (LED) respectively.

Unlike the light-emitting diode which produces or emits light, the photodiode is a photo detector which receives light and converts it into an electrical signal. A junction photodiode operates based on the photoelectric effect.

In total darkness, the diode junction has a relatively high internal resistance and therefore blocks current. However, when sufficient light energy strikes its semiconductor junction, the dynamic resistance of its junction decreases allowing more current to flow. Thus, photodiodes have a photoconductive gain.

Example No1

A junction photodiode is connected in reverse-bias mode across a 10 volt supply. What will be the diodes junction resistance if a reverse current of $2.0\mu\text{A}$ flows when exposed to a light intensity of 0.1 mW/cm^2 increasing to $50\mu\text{A}$ when the light source is increased to 20 mW/cm^2

1. Junction Resistance at 0.1 mW/cm^2

$$R_D = \frac{V_{PD}}{I_{PD}} = \frac{10\text{V}}{2.0\mu\text{A}} = 5\text{M}\Omega$$

2. Junction Resistance at 20 mW/cm^2

$$R_D = \frac{V_{PD}}{I_{PD}} = \frac{10\text{V}}{50\mu\text{A}} = 200\text{k}\Omega$$

Then we can see that a photodiodes junction’s dynamic resistance changes from $5\text{M}\Omega$ down to $200\text{k}\Omega$ as the light intensity increases. Therefore we could say that photodiodes can be thought of as variable-resistive devices controlled by light intensity.

Photodiode Construction

Basically, the photodiode consists of a combination of two different types of semiconductor materials to create a conductive junction between them. A single pn-junction is formed by doping silicon (or other semiconductors like

germanium, gallium arsenide, or indium gallium arsenide) with different types of impurities, depending on the wavelength of light to be detected.

This allows some photodiodes to respond to visible light, while others like the 5mm round [940nm Photodiode for Arduino](#) responds more to the upper infrared (IR) wavelength.

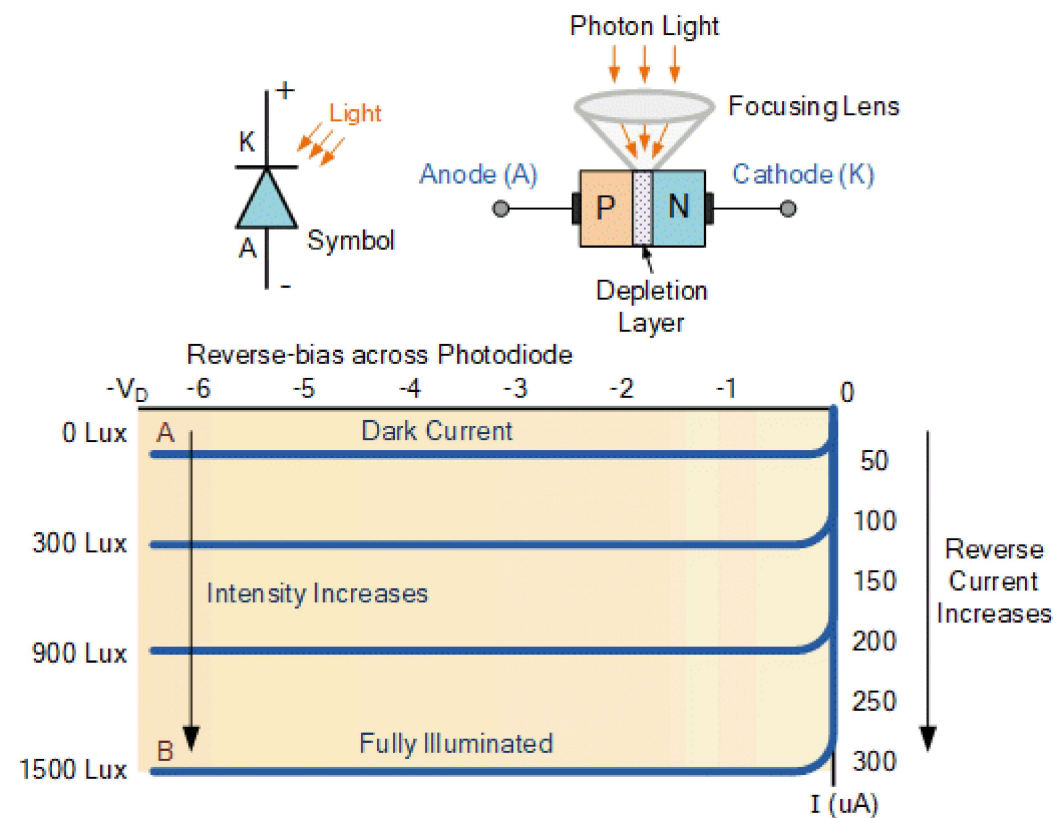
The p-region contains holes (positive charge carriers) while the n-region contains free electrons. That is, the n-type material has a surplus, while the p-type material has a deficiency of electrons. As a result, a depletion region is formed where the mobile charge carriers are depleted, creating an internal electric field across the pn-junction.

Electrons are swept from the p-region into the n-region, and holes in the opposite direction. Since the charge carriers are evenly separated across the junction between the two oppositely doped zones, a built-in voltage is established across the pn-junction.

When external photon light with sufficient energy strikes the photodiodes junction, it creates electron-hole pairs by exciting electrons from the valence band to the conduction band of the semiconductor material. If the pn-junction is connected, a small photocurrent proportional to the amount of incident light intensity will flow across the junction. This is known as the zero-bias **Photovoltaic effect**.

If an external reverse voltage is applied across the photodiode, it will widen the depletion region of the junction increasing its sensitivity to light. Thus, photodiodes are light sensitive diodes which operate in a reverse-bias condition (similar to Zener diodes) controlling current flow in direct proportion to the intensity, expressed as irradiance (mW/cm^2), of the light source as shown.

Basic PN-junction Photodiode and Symbol



Then we can see that a photodiode requires a constant reverse-biased excitation supply and for a given reverse voltage, current flow increases with an increase in light. Then from the photodiodes I-V characteristics curves above, at very low light intensity virtually no current flows, only a small leakage current. This is referred to as the photodiodes dark current. Note that a lower dark current indicates better performance, especially in low-light applications.

As the illumination level on the pn-junction increases, the resistance of the junction decreases and the reverse current rises, making the characteristic curve migrate towards point “B”. So as long as the reverse-bias voltage across the diode does not exceed its breakdown threshold, the photodiode behaves as a constant current generator, with the value of the current depending on the intensity of the incident light.

Note that the symbol of a photodiode is very similar to the normal pn-junction rectifying diode except that it contains arrows pointing towards the diode. These arrows represent light or photons of energy.

Types and Variations of a Photodiode

As well as the basic pn-junction photodiode described above, other types of photodiodes exist which can also be used as photo detectors. These include:

The PIN Photodiode

The **PIN photodiode** has an intrinsic (undoped) layer, (“i”) between its positive (P) and its negative (N) semiconductor regions. This creates a wider depletion region which allows the PIN photodiode to respond to much lower light levels across a much greater spectral bandwidth. Also, the fixed electric field in the wider “i” depletion region means there is a greater chance of free electrons being generated as less energy is required to penetrate deeper into the depletion region.

Then the *PIN photodiode* is more efficient than the standard pn-junction photodiode as this additional intrinsic layer reduces the diode junctions internal capacitance value, allowing for a faster response over a wide range of incandescent light. this makes PIN photodiodes ideal for high-frequency applications and fibre optic communications.

Avalanche Photodiodes

The **Avalanche photodiode** operates under high reverse bias voltage greater than 100 volts. This high voltage, low current bias causes highly excited free electrons to create additional electron-hole pairs through impact ionisation when subjected to light. That is, the avalanche photodiode exploits the phenomenon of avalanche multiplication as each photon of light causes multiple electrons to be released.

The advantage of the avalanche photodiode is this avalanche multiplication effect provides a high internal linear gain, dramatically increasing its sensitivity. Clearly the disadvantage is that they require a high voltage bias.

Schottky Photodiodes

The **Schottky photodiode** is based around the metal Schottky barrier diode. It use a metal-semiconductor junction instead of a normal pn-junction creating a built-in electric field at the interface. When light hits the semiconductor junction, photons generate electron-hole pairs with electrons moving toward the metal and holes toward the semiconductor, thus producing the required photocurrent.

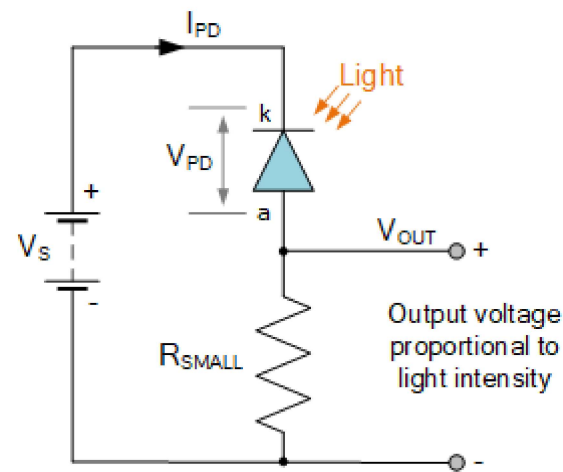
Schottky photodiodes offer faster spectral response times and good performance compared to the pn-junction, or the PIN photodiodes. This makes them ideal for high-speed and UV light detection applications due to their low capacitance and the absence of a depletion region in the metal semiconductor material.

How to Use A Photodiode

Now that we have learnt about the basic operation of the **Photodiode** and the different types available. Its time to see how we can use them in one of our circuits or projects.

The most basic photodiode light-sensing circuit is that of a voltage divider circuit as shown:

Simple Photodiode Circuit



Here, a photodiode such as the [BPW34 PIN Photodiode](#) is connected in reverse-bias to a constant-voltage excitation supply.

If the light intensity in mW/cm^2 is zero (darkness), only a very small reverse dark current will flow through the diode. Thus the I^2R voltage drop across the series resistor will be virtually zero, producing zero output voltage, V_{OUT} .

As the light level increases, photons strike the diodes pn-junction creating a photocurrent in the reverse-bias direction. This photocurrent produces a voltage drop, and therefore an output voltage across R . As the light intensity increases further, more current flows creating a bigger I^2R voltage drop.

The relationship between photocurrent and light intensity is based on the assumption that the reverse-bias voltage, (V_{PD}) across the diode is held constant. That means if we want the diode to behave like a current source which only depends on light intensity, then V_{PD} must be held nearly constant.

Therefore if we want the photodiode's voltage drop to remain constant as the photocurrent varies, the series resistance, R must be very small so that the voltage drop across it induced by the photocurrent is also small.

But if we want to sense the current produced by the photodiode, we will have to use some form of *current-to-voltage converter* which has a very small (ideally zero) input impedance.

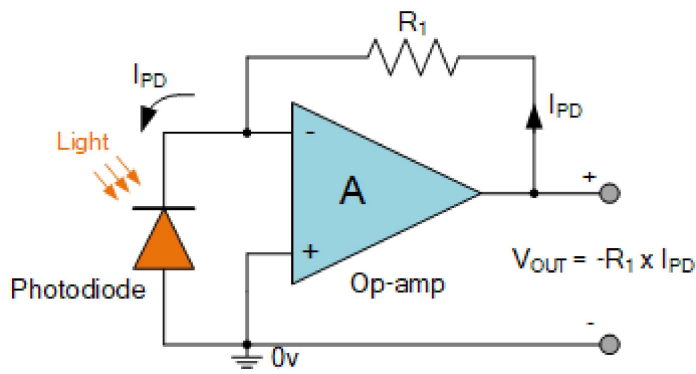
The Transimpedance Amplifier

The simplest and convenient way of transforming a current to a voltage is to use an operational amplifier configured as a *Transimpedance Amplifier*. Photodiodes can produce this current-to-voltage conversion when presented with a high impedance load but are typically interfaced with transimpedance amplifiers to convert their small photocurrent into a much larger voltage signal.

The transimpedance amplifier is a *current-to-voltage converter* (I-V converter), which as its name suggests, takes an input current, (I_{IN}) and outputs a voltage, (V_{OUT}). The gain of such an amplifier is basically: $V_{\text{OUT}} = A \cdot I_{\text{IN}}$. Where A is the gain of the amplifier in volts per ampere. That is: V/I or resistance, but is normally expressed in V/mA rather than ohms.

The advantage of using an op-amp as a photodiode amplifier is that we can make use of the op-amps virtual-earth input which makes the photodiode's load the amplifiers high input impedance without the need for the previous voltage divider circuit as shown.

Transimpedance Photodiode Amplifier



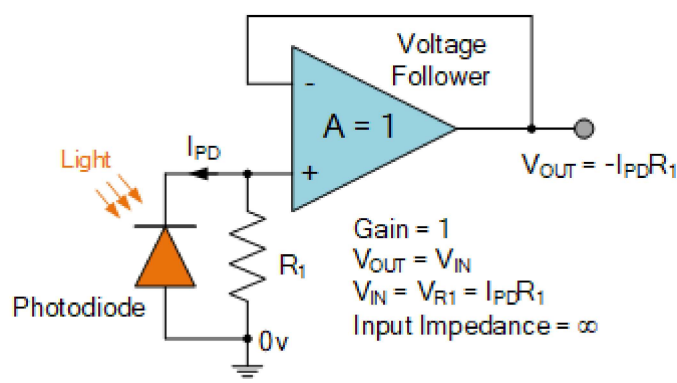
As we can see, the diode is connected to the virtual earth of the op-amp, with the photodiode current, I_{PD} being the same as the current through the feedback resistor, R_1 . This feedback forces the amplifier to convert the diode's photocurrent without any voltage being present at its inverting input.

The amplifiers open-loop gain measures the I_{PD} current through the photodiode and converts it into an output voltage per microampere input, (V/mA) equal to the diode current times the feedback resistance, R_1 . Thus the gain of the amplifier is determined by the resistive value of the feedback resistor R_1 . That is: $R_1 = A$.

Despite the above circuits simplicity, the current-to-voltage conversion of the transimpedance amplifier acting as a photodiode amplifier, can restrict its ac bandwidth due to the in-built parasitic capacitance of the diodes junction. This is because any voltage developed across the photodiode reacts with its junction capacitance, shunting away part of the diodes photocurrent at higher frequencies and reducing its upper bandwidth.

One way to overcome this issue and extend the ac bandwidth of the circuit is to isolate the photodiode completely from the signal voltage by configuring the op-amp as a unity-gain buffer as shown.

Voltage Follower Amplifier



Here, the photocurrent (I_{PD}) generated by the diode circulates through the resistor R_1 producing a voltage drop across it. Since the op-amp is configured as a voltage follower, it isolates the voltage developed across R_1 from its output delivering an almost ideal output voltage signal of: $V_{OUT} = -I_{PD} \cdot R_1$.

Example No2

A photodiode with a sensitivity of 5mV/pA is to be used with a transimpedance amplifier to construct a light meter. If the photodiode outputs a small current that is proportional to the incident light intensity between a range of 50pA @ 0.01 lux (dark current) to 5nA @ 1000 lux (full brightness). Calculate the value of feedback resistor required for the amplifier.

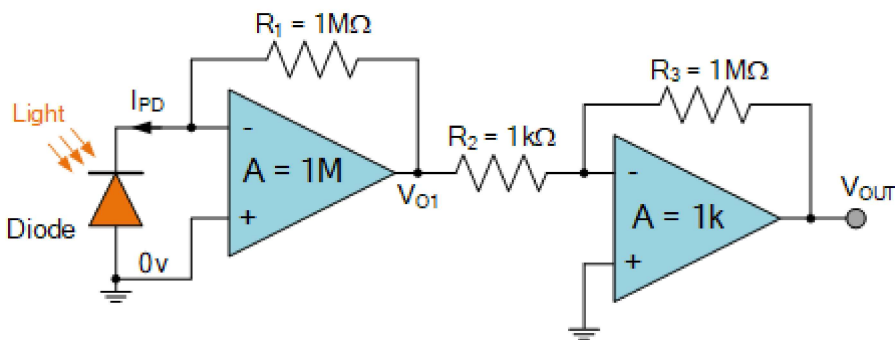
If the sensitivity of the diode is 5mV/pA and it produces a photocurrent of 5nA at 1000 lux. Then:

$$R_1 = 5\text{mV} \times \frac{1000}{5 \times 10^{-9}} = 1 \times 10^9 = 1000 \text{ M}\Omega$$

This means that the overall gain of the op-amp would be: 1×10^9 . While it would be possible to implement such a high gain using one single-stage amplifier circuit with a $1000 \text{ M}\Omega$ feedback resistor. Any amount of noise or instability within the circuit or power supply no matter how small would also be amplified 1×10^9 times.

Another approach would be to reduce the gain of the transimpedance amplifier to produce an output voltage in nano-volts, (nV) and then use a second [inverting operational amplifier](#) to amplify the signal up to the required mill-volts, (mV) output as shown.

Photodiode Amplifier Circuit



Here, we have selected the transimpedance amplifier feedback resistor to a more appropriate value of $1 \text{ M}\Omega$ which will produce a voltage gain for the first stage of 1×10^6 . That is 1000 nV/pA . If the overall gain required is 1×10^9 , then the voltage gain of the second stage will be: 1×10^3 . That is 1000 .

Thus, the voltage output of the circuit at 5pA (dark current) will be:

$$50 \text{ pA} \times 1 \times 10^9 = 50 \text{ mV}$$

and the voltage output of the circuit at 5nA (full brightness) will be:

$$50 \text{ nA} \times 1 \times 10^9 = 5 \text{ Volts}$$

Tutorial Summary

We have seen here that *Optoelectronic devices* either produce light or use light in their operation. The versatility and reliability of photodiodes make them indispensable in modern optoelectronic systems since they can be used to convert light energy (photon energy) directly into electric current.

Photodiodes are two-lead semiconductor devices used in their photovoltaic mode with zero bias to provide a linear operation and high switching speeds. A small amount of current called dark current will flow through the diode even when there is no illumination. However, when a photodiode is illuminated, it allows a small current to flow creating a voltage.

Photodiodes come in all shapes and sizes along with built-in lenses or optical filters to detect different wavelengths of light. As such are widely used in imaging applications with photodiode arrays forming the basis of digital cameras and scientific instruments, ambient light sensing in displays, and safety systems like smoke detectors.

Read more Tutorials in Diodes

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- [3. PN Junction Diode](#)
- [4. The Signal Diode](#)
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